

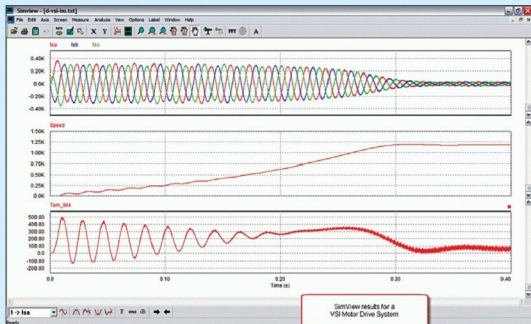


Fig. 2: Display of simulation results using Simview

System requirements

PSIM runs on Microsoft Windows 7 or 8, either on a standalone personal computer (PC) or on a network PC. It is available in 32-bit or 64-bit architecture.

A minimum of 500MB RAM memory is required. The program will occupy around 250MB hard disk space after installation.

Easy modelling. There are a lot of modelling options available in PSIM. With the help of some add-on modules, designers can simulate the control circuitry with op-amps, s-domain blocks, in the discrete z-domain or with the embedded runtime C compiler. The frequency analysis of a topology preserves the switching behaviour; there are non-average models used, so non-linear aspects can be accurately measured.

Also, there are a lot of application-specific add-on modules for the model. By using modules for motor drives, digital signal processor (DSP) and field programmable gate array (FPGA) support and renewable energy, we can adapt the models to suit various applications. PSIM can also handle magnetic simulation where users can build custom magnetic circuits with non-linear cores that

exhibit hysteresis.

Co-simulation with other environments. SimCoupler module can facilitate co-simulation by providing an interface between PSIM and MATLAB/Simulink. We can thus implement a part of the system using PSIM and the rest using MATLAB/Simulink.

The set up and simulation time of systems is greatly reduced by allowing MATLAB/Simulink users to implement and simulate power circuits in their original circuit form using SimCoupler module. Power electronic researchers can also benefit from this facility by simulating the control in MATLAB/Simulink environment.

PSIM can co-simulate with ModelSim, using PSIM's ModCoupler module, for VHDL and Verilog code verification. If a user defines a control scheme with z-domain blocks, PSIM's SimCoder module can translate that algorithm into C code, and with a few additional blocks one can also generate a code that will run directly on a supported DSP with no additional coding steps.

What is new in PSIM 10

A lot of new features have been included in the latest version of PSIM.

Let us go through some of these.

Diode model with reverse recovery. PSIM 10 comes with a diode model with reverse recovery. The designer can simulate the diode turn-off process and help predict switching phenomena caused by the diode reverse recovery using this model.

Also, there is provision for the MOSFET model with turn-on/turn-off switching transient. With the help of this model, a developer

can easily simulate the switching transient and design the gate drive circuit.

New multisine AC sweep method.

This version of the software has adopted a new AC sweep method. It is based on multisine algorithm. In majority of cases, this method gives excellent results in comparison with earlier methods.

Multi-level modelling. The developer might require different levels of models at various stages of simulation. The multi-level modelling feature incorporated in the latest version of this software allows the user to easily switch from one level of model to another by just changing a flag, without changing the schematic. This is a great advantage for the simulation process.

Inclusion of IC models. In order to facilitate the user, a lot of IC models have been included to the standard PSIM library. These include 555 timer, UC3823, UC3825, UC3842/3843, UC3844/3845, UC3846, UC3854, UC3872, UCC3806, UCC3817/3818, UCC3895 and UCC28070.

Besides, there are many interesting features like the ultracapacitor model in the renewable energy module, two solar cell models for crystal-